

Day 2 Assignment – Dictionaries, Sets & Functions

Intern Coding Challenge Sheet

Instructions

- Write all programs in Python.
 - Use proper variable names.
 - Add comments wherever needed.
 - Write clean and readable code.
 - Practice all questions by yourself.
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Section 1: Dictionary Challenges

Q1.

Create a dictionary to store:

- Student name
- Age
- Course
- City

Print all dictionary values.

Q2.

Create a dictionary of employee details and access:

- Employee name
- Salary
- Department

using keys.

Q3.

Add a new key-value pair to an existing dictionary.

Example:

- Add `"email"` to student details.
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Q4.

Update the value of an existing key in a dictionary.

Q5.

Remove a key from a dictionary using:

- `pop()`
 - `del`
-

Q6.

Print all:

- Keys
- Values
- Key-value pairs

from a dictionary using loops.

Q7.

Count how many subjects are present in a student marks dictionary.

Q8.

Create a dictionary containing marks of 5 subjects and calculate the total marks.

Q9.

Check whether a key exists in a dictionary or not.

Q10.

Create a dictionary of products with prices and display only products having price greater than 500.

Section 2: Set Challenges

Q11.

Create a set containing 5 numbers and print all elements.

Q12.

Add new elements into a set using `add()`.

Q13.

Remove an element from a set using:

- `remove()`
 - `discard()`
-

Q14.

Create two sets and perform:

- Union
- Intersection
- Difference

operations.

Q15.

Find common elements between two sets.

Q16.

Check whether a particular value exists in a set.

Q17.

Take a list containing duplicate values and convert it into a set.

Q18.

Create a set of student names and display the total number of unique students.

Q19.

Create two sets of numbers and check whether both sets are equal or not.

Q20.

Remove all elements from a set using `clear()`.

Section 3: Function Challenges (Basic Level)

Q21.

Create a function that prints:

"Welcome to Python Programming"

Q22.

Create a function to add two numbers and display the result.

Q23.

Create a function that accepts a name as a parameter and prints:

"Hello, Rohit"

Q24.

Create a function to find the square of a number.

Q25.

Write a function to check whether a number is even or odd.

Q26.

Create a function to calculate the area of a rectangle.

Q27.

Write a function that returns the greater number between two numbers.

Q28.

Create a function to print numbers from 1 to 10.

Q29.

Write a function to calculate the sum of all numbers in a list.

Q30.

Create a function that checks whether a student is pass or fail based on marks.

Condition:

- Pass → marks ≥ 35
 - Fail → marks < 35
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Q31.

Create a function with default arguments.

Example:

- Default city should be "Pune"
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Q32.

Create a function using keyword arguments.

Q33.

Write a function that accepts multiple numbers using `*args` and prints them.

Q34.

Create a function that returns:

- Maximum number
- Minimum number

from a list.

Q35.

Write a function to count vowels in a string.

Best of Luck Interns

Practice daily to improve your Python programming skills.